Md Shahanur Islam Shagor

Full-Stack Developer & AI Engineer

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I am writing to apply for the Simulationsingenieur Elektrotechnik position at MY Humancapital GmbH. My work in software engineering and autonomous systems has given me practical experience with simulation, sensor integration, and system-level analysis that aligns with the responsibilities of this role.

Through research and development work, I have built real-time object detection systems using YOLOv8n on edge hardware such as the NVIDIA Jetson Nano and Raspberry Pi. These projects required modeling environmental interactions, optimizing inference latency, and analyzing thermal and computational constraints on embedded platforms. I also developed a custom UDP-based streaming protocol to achieve ultra-low latency communication between camera modules and processing units, which involved simulating network behavior and tuning for reliability under real-world conditions.

One area that particularly interests me is the development of Digital Twin approaches for vehicle systems. In my work on OmniGuard V2X, a privacy-preserving blockchain framework for autonomous vehicle networks, I designed and simulated secure communication protocols for Vehicle-to-Everything (V2X) environments. This required modeling data flows, evaluating cryptographic overhead, and testing system resilience against cyber and physical threats. The experience taught me how to bridge simulation models with practical hardware constraints.

In addition to my technical experience, I work effectively across multidisciplinary teams and communicate clearly with both technical and non-technical stakeholders. I am fluent in Bengali and English, have working knowledge of Russian and Hindi, and enjoy collaborating in international and cross-cultural environments.

I have attached my CV for your review. I welcome the opportunity to discuss how my background in simulation, embedded systems, and autonomous technologies can contribute to your team.

Sincerely,

Md Shahanur Islam Shagor
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